

The art of information



Class 9: 01/15 (Language)



Word lengths are optimized for efficient communication

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We demonstrate a substantial improvement on one of the most celebrated empirical laws in the study of language, Zipf's 75-y-old theory that word length is primarily determined by frequency of use. In accord with rational theories of communication, we show across 10 languages that average information content is a much better predictor of word length than frequency. This indicates that human lexicons are efficiently structured for communication by taking into account interword statistical dependencies. Lexical systems result from an optimization of communicative pressures, coding meanings efficiently given the complex statistics of natural language use.

information theory | rational analysis

One widely known and apparently universal property of human language is that frequent words tend to be short. This law was popularized by Harvard linguist George Kingsley Zipf, who observed that “the magnitude of words tends, on the whole,

(11, 6): the amount of information conveyed by a word should be linearly related to the amount of time it takes to produce—approximately, its length—to convey the same amount of information in each unit of time. A constant information rate can make optimal use of the speech channel by maximizing the amount of information conveyed, without exceeding the channel capacity of speech or our cognitive systems (12, 13). Thus, lexical systems that assign length according to information content can be communicatively more efficient than those that use frequency.

Importantly, the amount of information conveyed by an instance of a word depends on its context. To formalize this, we can consider two random variables, C for contexts and W for words, with a joint distribution $P(C, W)$ given by the natural statistics of language use. The average amount of information conveyed by a particular word w is given by the following (14):

$$-\sum_c P(C = c | W = w) \log P(W = w | C = c). \quad [1]$$

“In addition, there are several possible ways to measure word length. Here, we primarily use orthographic length because it is readily available from corpora and tends to be highly correlated with both phonetic length and production time”

n-grams

language modeling

how do we make
something that
generates text?

how do we predict the
next word?

Typing with next

Aa



+

word

time

speed



q

w

e

r

t

y

u

i

o

p

a

s

d

f

g

h

j

k

l



z

x

c

v

b

n

m



123



space

return



N = 1 : This is a Sentence unigrams: this,
is,
a,
sentence

N = 2 : This is a Sentence bigrams: this is,
is a,
a sentence,

N = 3 : This is a Sentence trigrams: this is a,
is a sentence,

i visited the city of new york

of _____

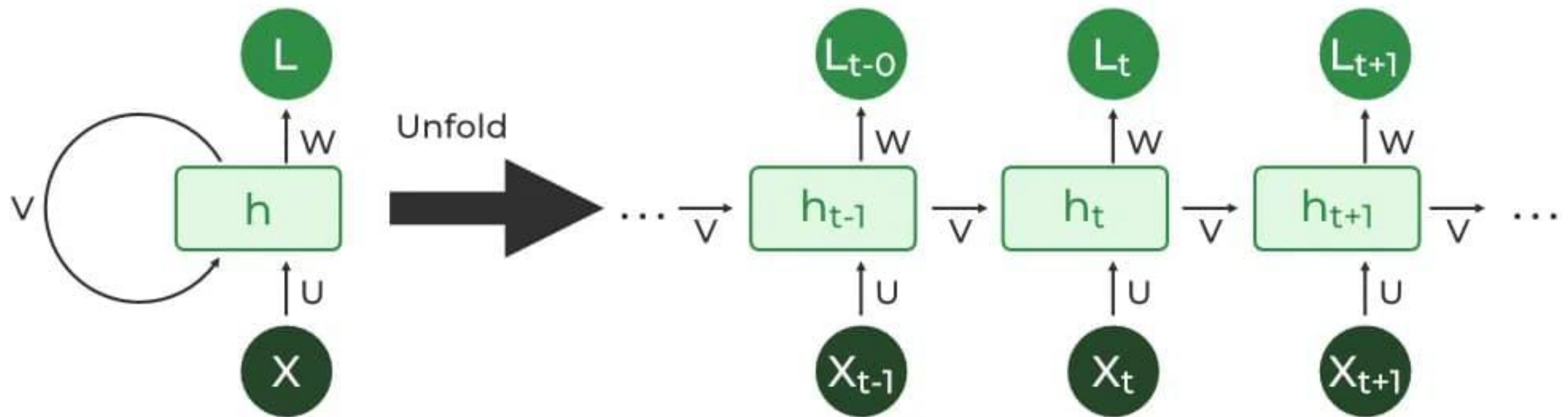
“Zipf’s idea of assigning word length by frequency can be maximally concise and efficient if words occur independently from a stationary distribution. However, natural language use is highly nonstationary as word probabilities change depending on their context”

but...

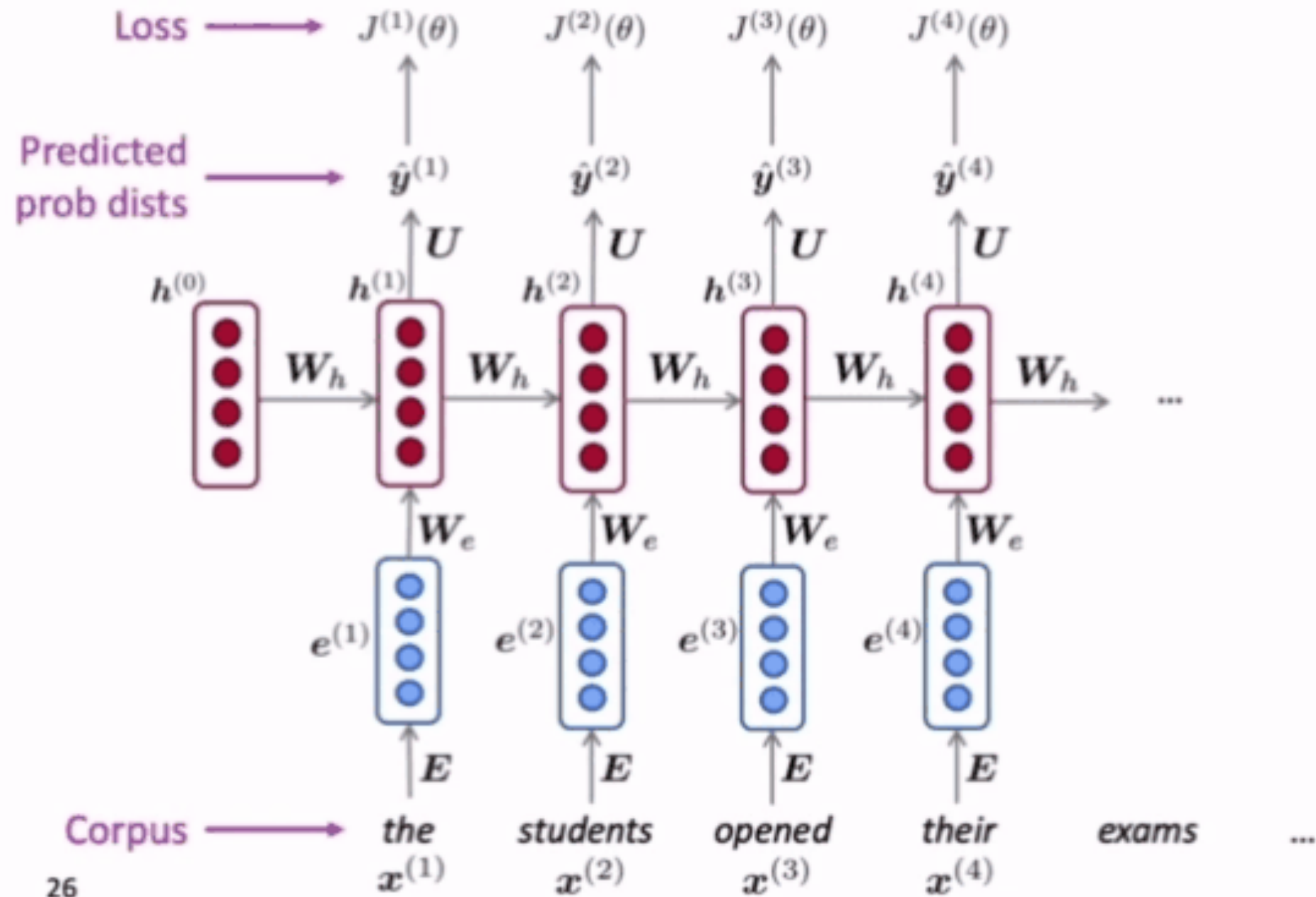
In *Harry Potter*, the students play a game called every year. In this game, people ride _____

?????

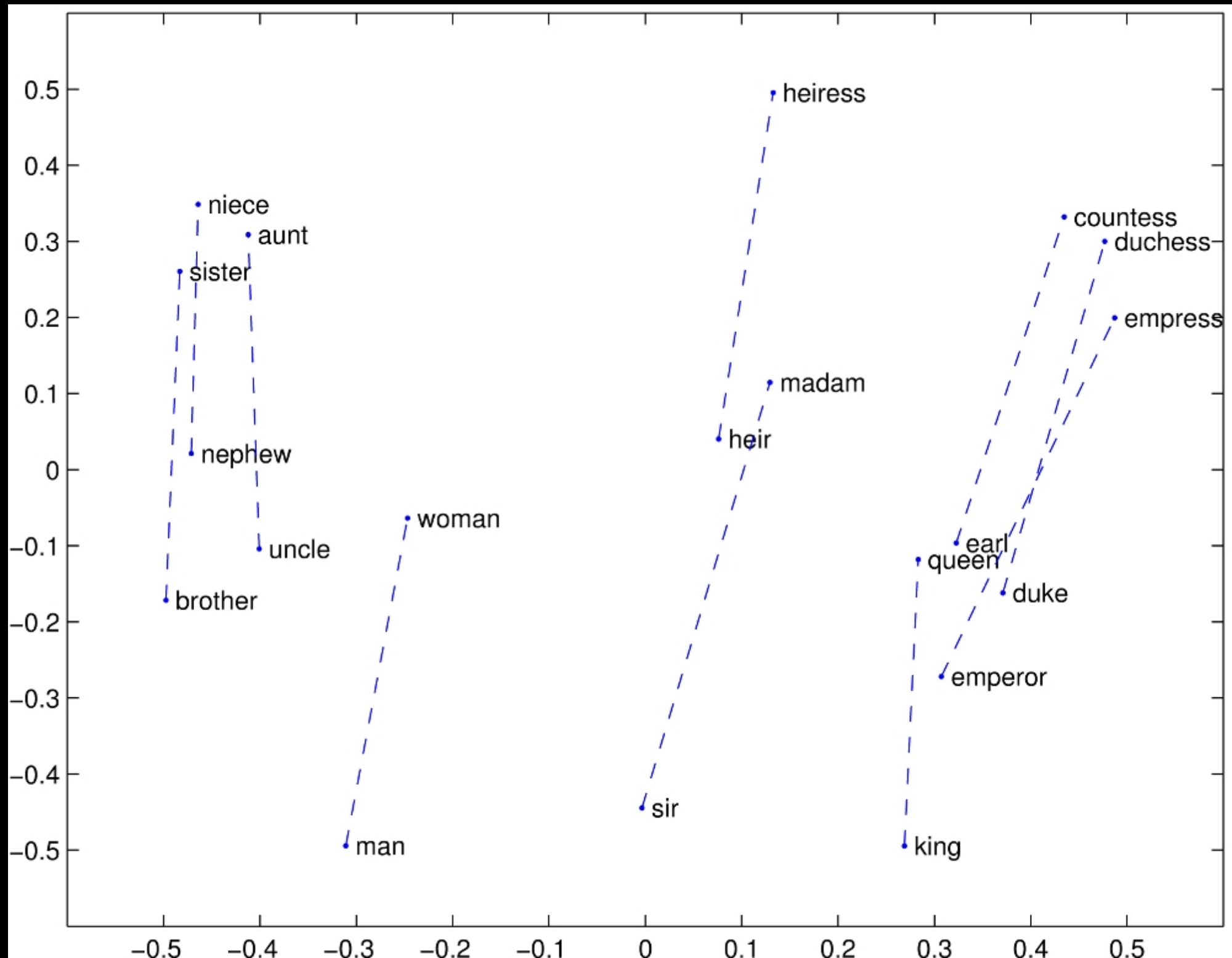
RNN



Training a RNN Language Model



word embeddings



but...

extremely slow
(can't be parallelized)

Attention Is All You Need

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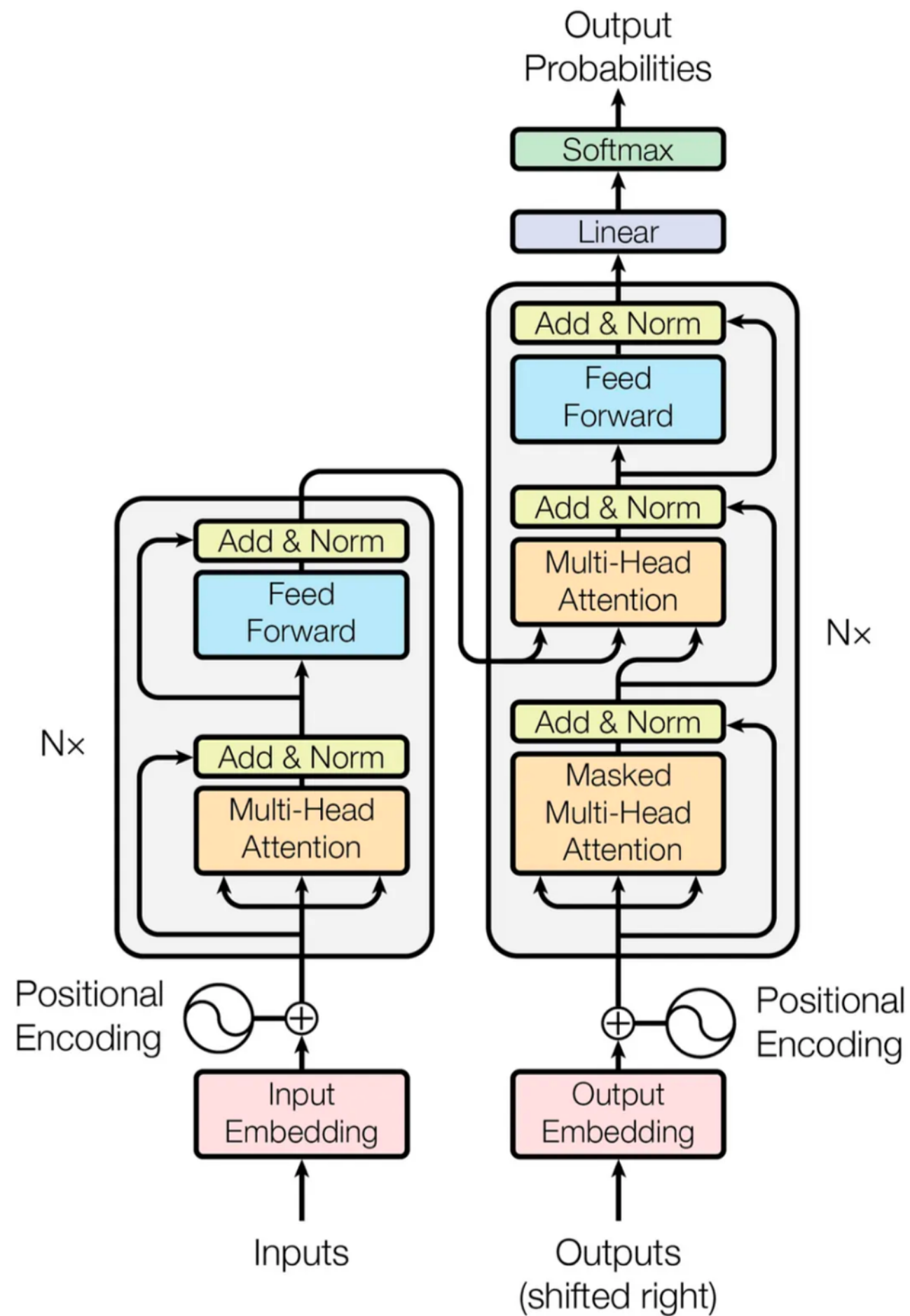
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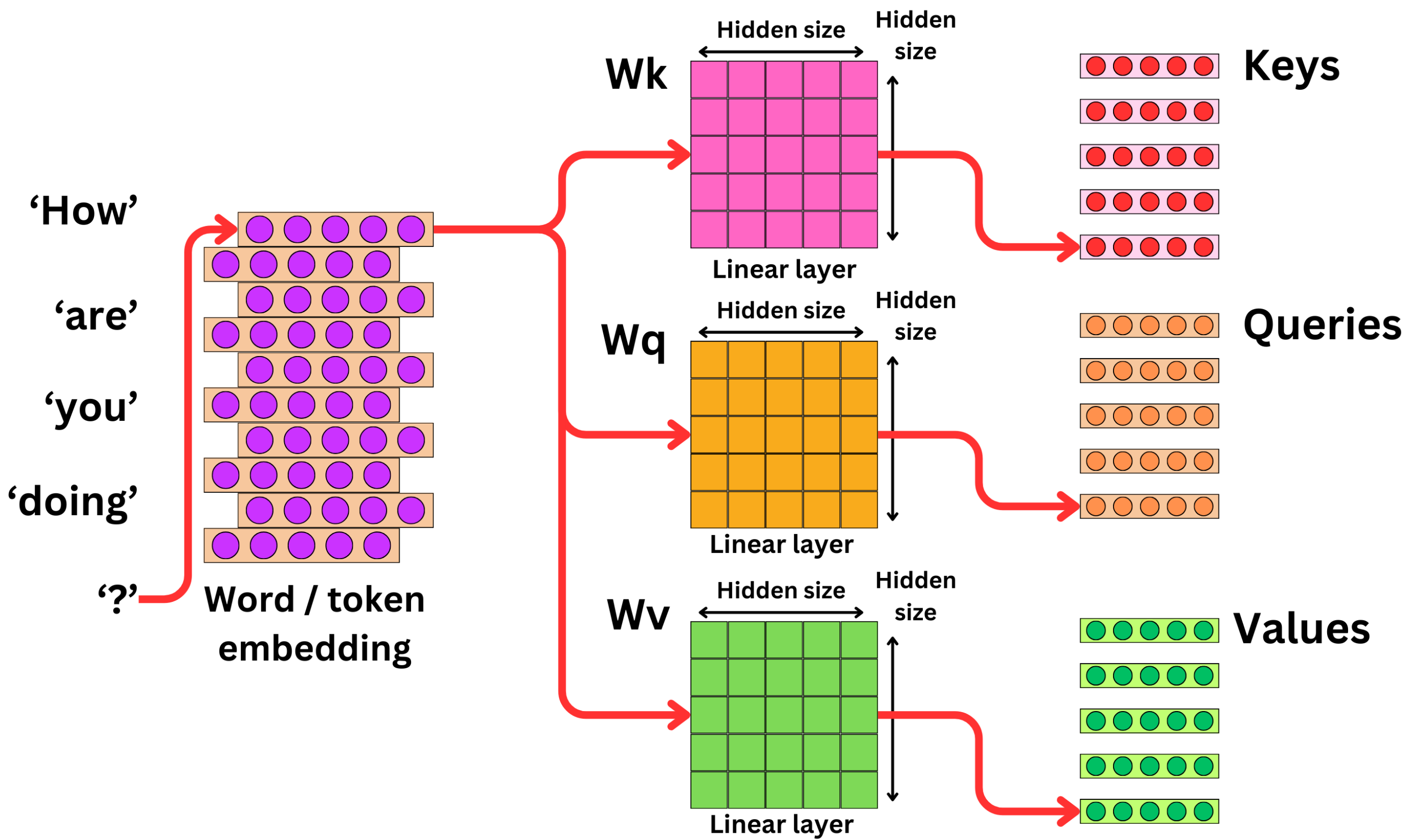
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Abstract

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results, including ensembles, by over 2 BLEU. On the WMT 2014 English-to-French translation task, our model establishes a new single-model state-of-the-art BLEU score of 41.0 after training for 3.5 days on eight GPUs, a small fraction of the training costs of the best models from the literature.

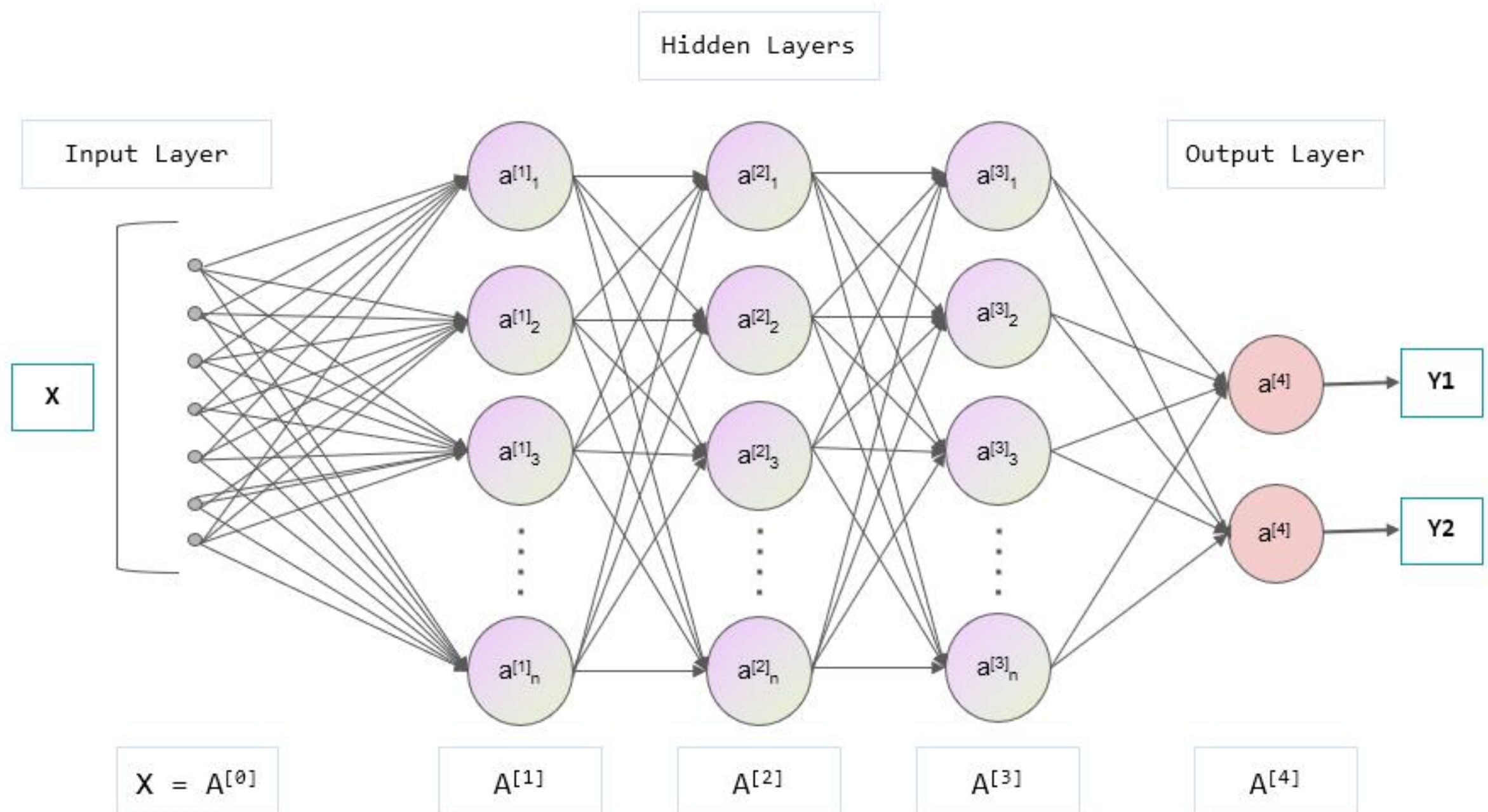








BERT



ARTIFICIAL PHANTASIA: EVIDENCE FOR PROPOSITIONAL REASONING-BASED MENTAL IMAGERY IN LARGE LANGUAGE MODELS

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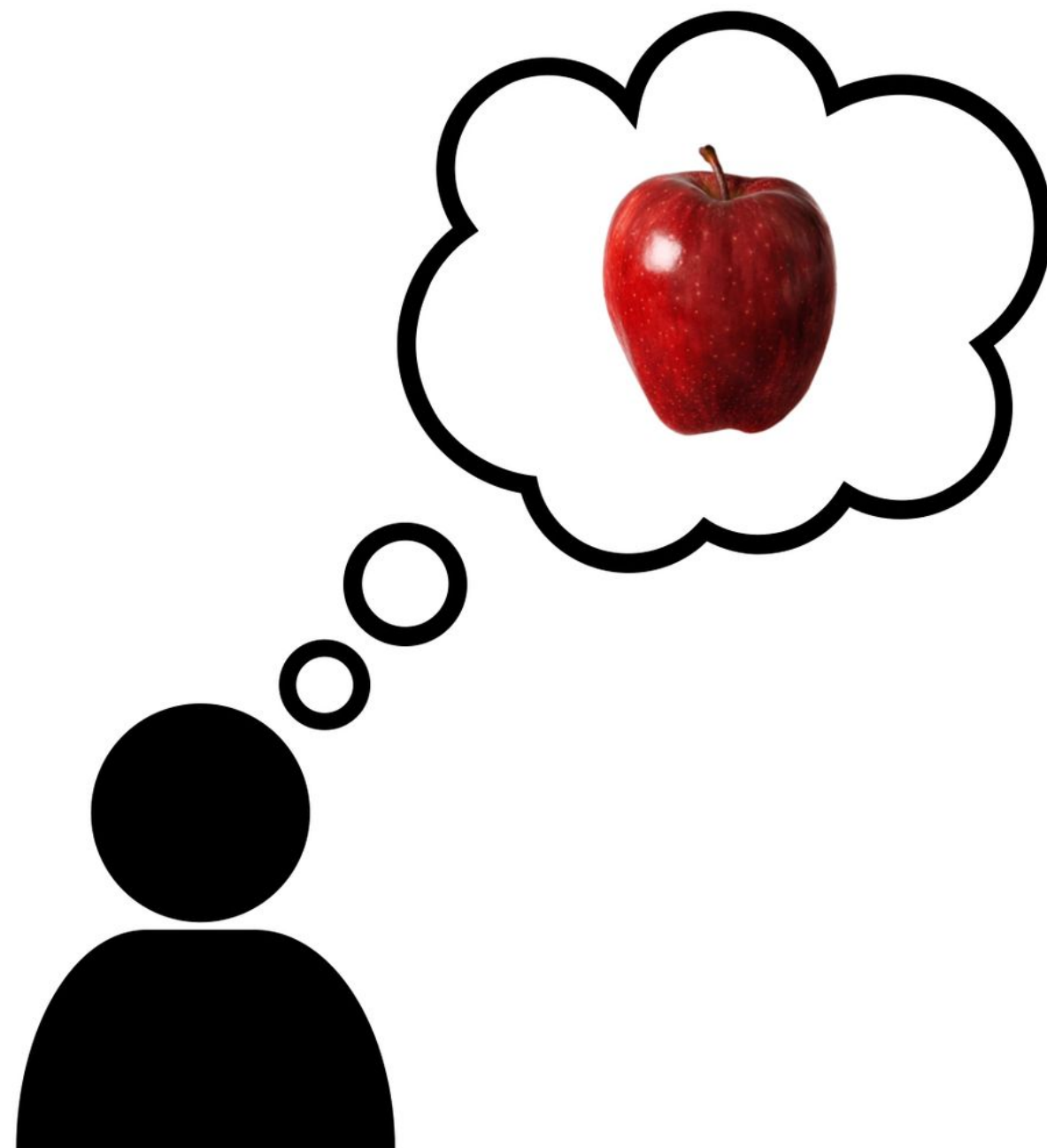
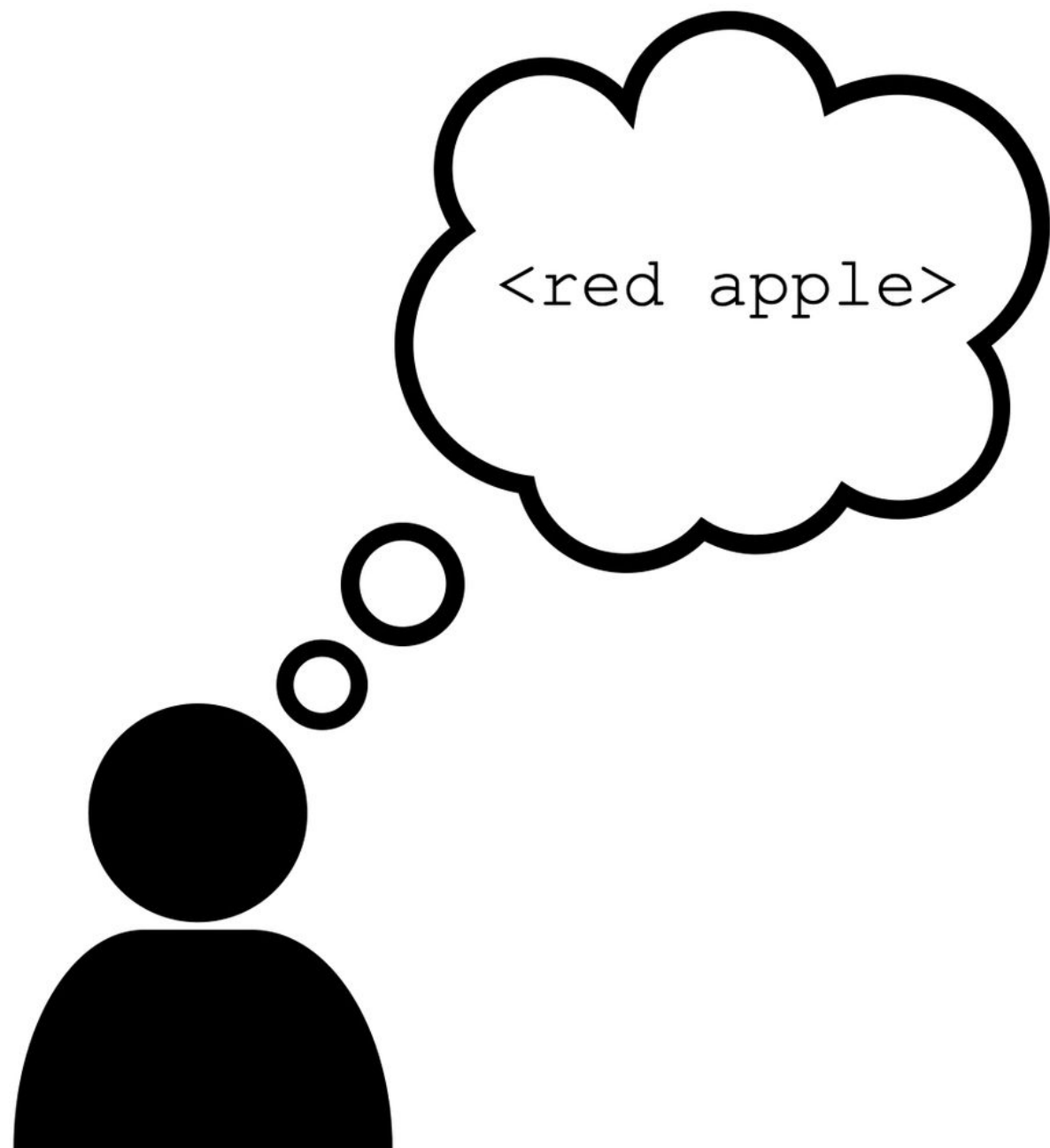
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ABSTRACT

This study offers a novel approach for benchmarking complex cognitive behavior in artificial systems. Almost universally, Large Language Models (LLMs) perform best on tasks which may be included in their training data and can be accomplished solely using natural language, limiting our understanding of their emergent sophisticated cognitive capacities. In this work, we created dozens of novel items of a classic mental imagery task from cognitive psychology. The task consists of following a series of short instructions (3-5 steps), performing basic transformations on imagined letters and simple shapes to create a mental image of an object, and finally recognizing and labeling the object. Traditionally, cognitive psychologists have argued that this task is solvable exclusively via visual mental imagery (i.e., language alone would be insufficient). LLMs are perfect for testing this hypothesis. First, we tested several state-of-the-art LLMs by giving text-only models written instructions and asking them to report the resulting object after performing the transformations in the aforementioned task. Then, we created a baseline by testing 100 human subjects in exactly the same task. We found that the best LLMs performed significantly above average human performance (9.4%-12.2% increase over the human average of 54.7%, $p < .00001$). Finally, we tested reasoning models set to different levels of reasoning and found





```
> imagine a red apple
```

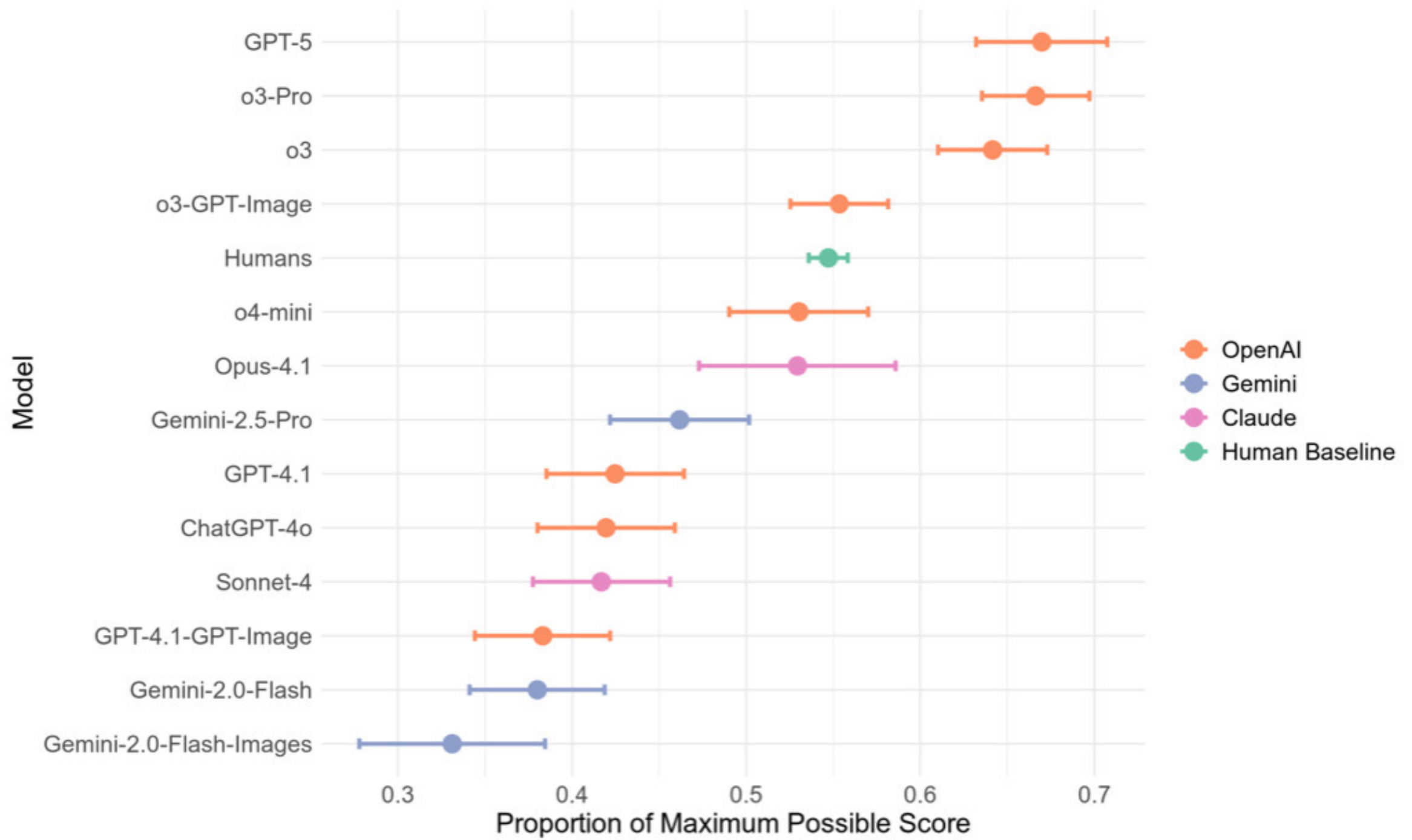
```
> [0.1325, 0.8472, 0.4561,  
0.9934, 0.2739, 0.6510, 0.3184,  
0.7527, 0.0941, 0.5028, 0.6417,  
0.2376, 0.8293, 0.4108, 0.5649,  
0.3821, 0.7154, 0.1987, 0.9342,  
0.2896, 0.7154, 0.1987, 0.4782,  
0.1269, 0.8035, 0.6674, 0.3518,  
0.9207, 0.1584, 0.4872, 0.7613,  
0.3069, 0.6984, 0.4237, 0.5791,  
0.2438, 0.8365, 0.1927, 0.9148,  
0.5082, 0.6273, 0.3826, 0.7431,  
0.2719, 0.8194, 0.3567, 0.9052,  
0.1894, 0.6698, 0.2481, 0.7842,  
0.3327, 0.9275, 0.4926, 0.6158,  
0.3851, 0.7543, 0.2148, 0.8437,  
0.1274, 0.9382, 0.2785, ...]
```


	Step 1	Step 2	Step 3
Instructions	Imagine a capital letter "D".	From there, imagine the figure rotated 90 degrees to the left.	From there, imagine a capital letter "J" attached to the bottom center of the figure.
Mental Image	D		

Answer: Umbrella

	Step 1	Step 2	Step 3	Step 4
Instructions	Imagine two lowercase letter “o” next to each other.	From there, imagine the two figures pressed together so that they are just touching.	From there, imagine another lowercase letter “o” just touching the top of the figure in the center.	From there, imagine a lowercase letter “v” just touching the bottom of the figure in the center.
Mental Image				

Answer: Ice cream, flower bouquet, balloons



tomorrow



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